2024 Calculus II Supplement Homework (2)

1. Supplement Homework, Exercise 1,2,4,9,10, p.24-25.

2. Which of the following series converge absolutely, which converge conditionally, and which diverge ? Give your reasons.

(a) $\sum_{n=1}^{\infty}(-1)^n\, 2^n \sin\frac{\pi}{3^n}$. (b) $\sum_{n=1}^{\infty}\left(\frac{\sin n}{n^2}+\frac{(-1)^n}{n}\right)$. (c) $\sum_{n=1}^{\infty}\frac{n!}{n^n}$. (d) $\sum_{n=1}^{\infty}\frac{(\ln n)^2}{n}$.

3. Supplement Homework, Exercise 19, p.25.

4. Supplement Homework, Exercise 25, p.26.

5. Supplement Homework, Exercise 35, p.27.